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MODEL SELECTION AND VALIDATION REPORT ( 2004_2011_biomass model )
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## OBSERVATIONS:1696

## SANITY: OK − all models in the selection process passed sanity checks.

## SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

## MODEL SUMMARY ( 2004_2011_biomass model )

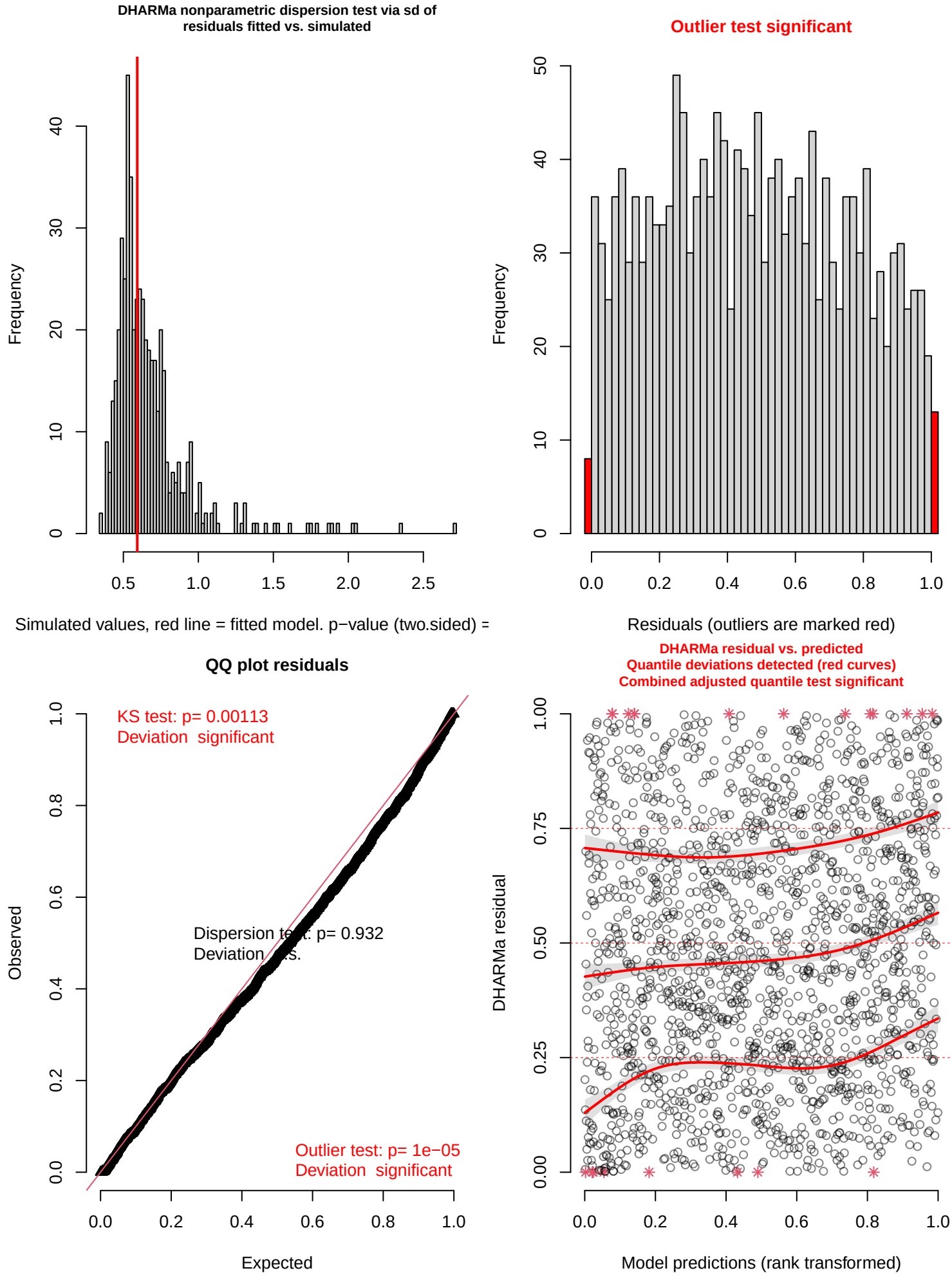
Spatiotemporal model fit by REML ['sdmTMB']
Formula: fixed_formula
Time column: year
Family: lognormal(link = 'log')

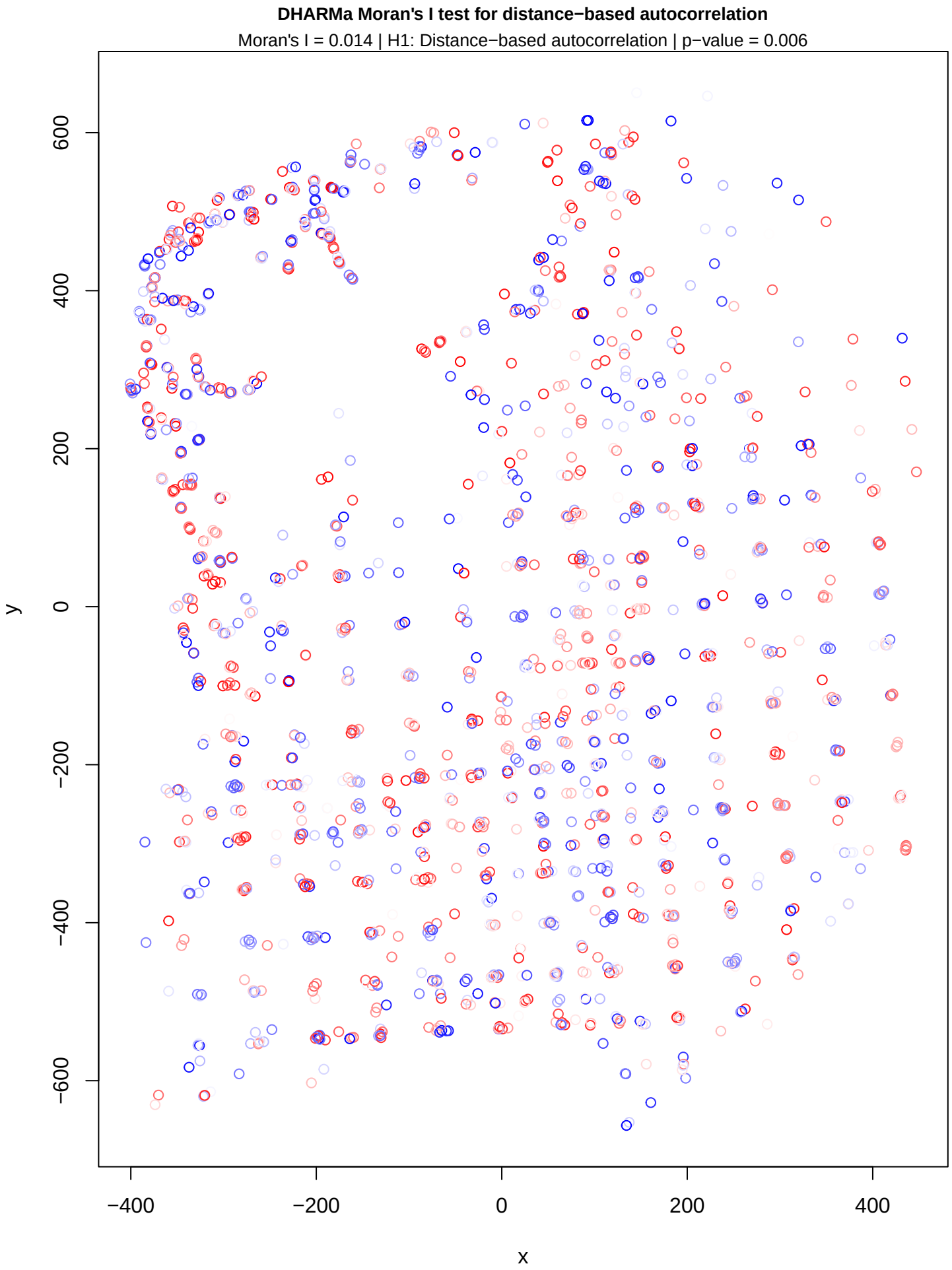
Conditional model:
              coef.est coef.se
(Intercept)       6.77    0.31
kde_fric          0.10    0.03
kde_feve         −0.21    0.03
tric              0.07    0.02
teve             −0.36    0.03
sbt_mean          0.27    0.07
sbt_sd            0.20    0.05
sst_sd            0.03    0.05
sic_mean          0.19    0.09
log_chla_mean    −0.03    0.03
depth             0.09    0.03

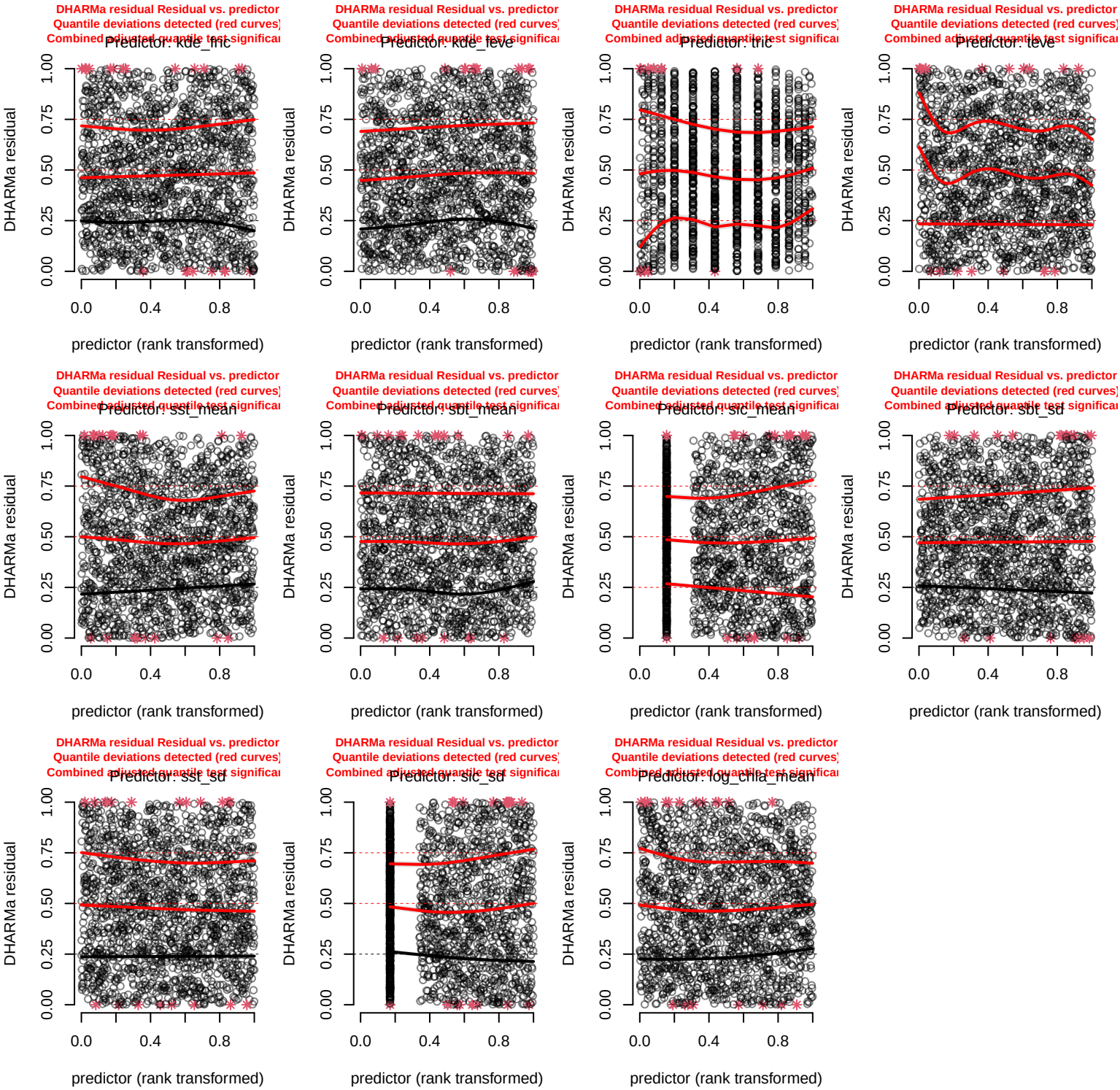
Dispersion parameter: 0.76
Spatiotemporal AR1 correlation (rho): 0.90
Matérn range: 371.26
Spatiotemporal marginal AR1 SD: 0.92
REML criterion at convergence: 13286.853

See ?tidy.sdmTMB to extract these values as a data frame.

## SELECTION STATISTICS
                                                   df      AIC  delta_AIC
spatiotemporal_random_fields_ar1                   15 26603.71   3.929905
spatiotemporal_random_fields_ar1_with_anisotropy 17 26607.64 147.369719
spatiotemporal_random_fields_iid                   14 26755.01   3.816102
spatiotemporal_random_fields_iid_with_anisotropy 16 26758.82  74.447763
spatial_random_fields                              14 26833.27   2.745211
spatial_random_fields_with_anisotropy              16 26836.02 585.121226
error_only                                         12 27421.14         NA
```


DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated
Frequency
Simulated values, red line = fitted model. p−value (two.sided) =
Outlier test significant
Frequency
Residuals (outliers are marked red)
QQ plot residuals
KS test: p= 0.00113
Deviation significant
Dispersion test: p= 0.932
Deviation n.s.
Outlier test: p= 1e−05
Deviation significant
Observed
Expected
DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
Model predictions (rank transformed)


DHARMa Moran's I test for distance−based autocorrelation
Moran's I = 0.014 | H1: Distance−based autocorrelation | p−value = 0.006
y
600
400
200
0
−200
−400
−600
x
−400
−200
0
200
400


DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significa
Predictor: kde_fnc
Predictor: kde_feve
Predictor: tric
Predictor: teve
Predictor: sst_mean
Predictor: sbt_mean
Predictor: sic_mean
Predictor: sbt_sd
Predictor: sst_sd
Predictor: sic_sd
Predictor: log_chla_mean
DHARMa residual
predictor (rank transformed)
0.00
0.25
0.50
0.75
1.00
0.0
0.4
0.8